AP Physics 1

Name _____

Period _____ Date _____

Practice Test – Introductory Dynamics (Version 4)

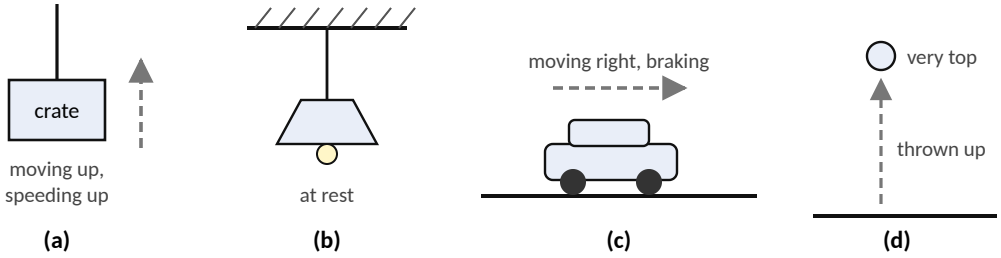
On any problems, you must show all work, include units, round properly, and box your final answer. Do the algebra first, before plugging in values.

Rewrite the following statement in the space below: On my honor, before God, I have neither given, nor received, any aid on this test.

For all questions:

- Use g = 9.8 m/s² (9.8 N/kg). Air resistance is negligible unless the question says it is not.
- Free-body diagrams: draw the forces themselves, not components, each as an arrow starting on the dot.
- Round answers to the significant figures of the given values.
- 50 minutes. Four questions, 25 points each (100 points total).

1. (25 points) For each situation, draw and label the forces (not components) exerted on the object named. Draw each force as its own arrow starting on the dot and pointing away from it, with lengths that show which forces are larger.

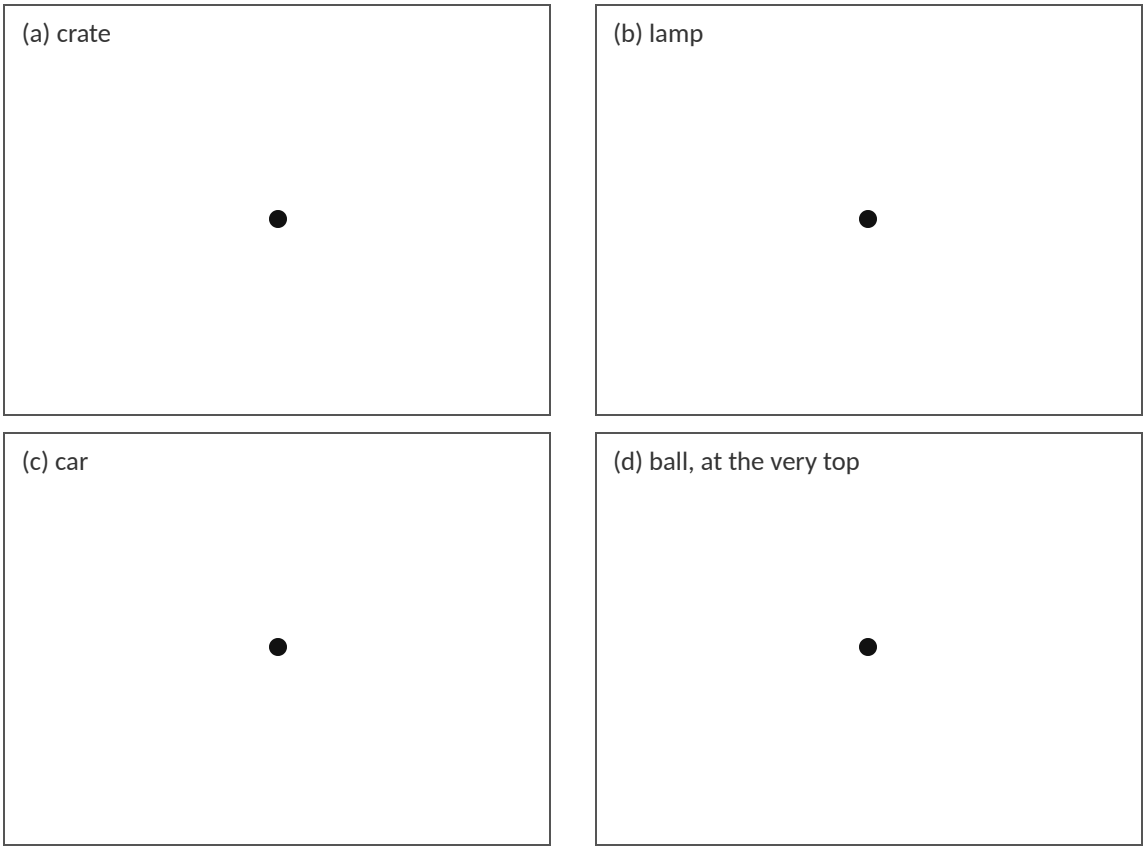


(a) A crane cable lifts a crate straight up. The crate is moving up and speeding up. Draw the free-body diagram of the crate.

(b) A lamp hangs at rest from a single cord attached to the ceiling. Draw the free-body diagram of the lamp.

(c) A car moves to the right on a level road while the driver brakes, so the car is slowing down. Draw the free-body diagram of the car.

(d) A ball is thrown straight up. Draw the free-body diagram of the ball at the very top of its flight. Air resistance is negligible.



(d) *continued*. Just before this instant, the ball was moving up. In one or two sentences, explain why there is no upward force on the ball.

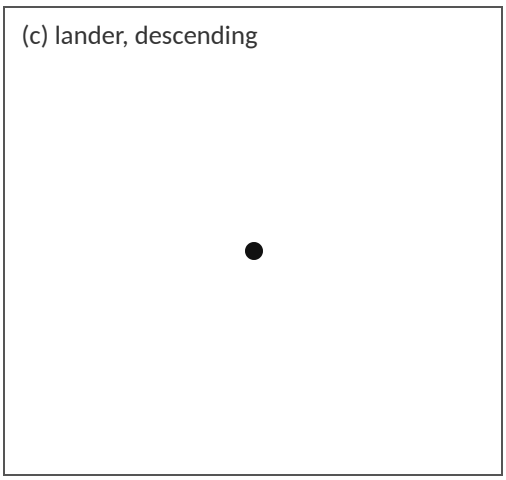
2. (25 points) A rover has a mass of 15.0 kg when it is measured on Earth. On the surface of Mars, the rover weighs 55.7 N.



(a) Calculate the gravitational field strength g on Mars, with units.

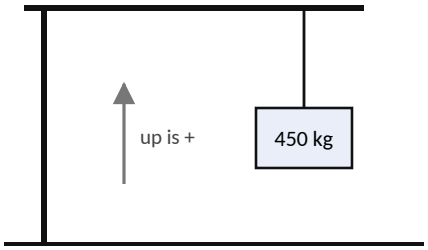
(b) What is the rover's mass on Mars? Explain why it is or is not different from its mass on Earth.

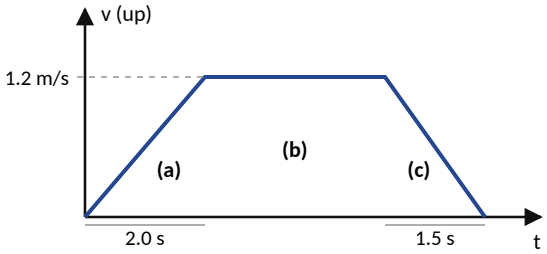
(c) A 6.20×10^2 kg lander descends straight down toward the surface of Mars. Its engine pushes exhaust gas downward, and the lander slows down at a rate of 1.50 m/s². Ignore the thin Martian air. Draw the free-body diagram of the lander, then calculate the magnitude of the thrust force on it. Use your value of g from part (a).



(d) Identify the Newton's third law partner of the thrust force: what exerts it, on what object, its magnitude, and its direction.

3. (25 points) A crane lifts a 450 kg crate straight up from rest. The cable is the only thing pulling up on the crate, and air resistance is negligible. Take up as positive. Start each part from a quick free-body diagram of the crate.





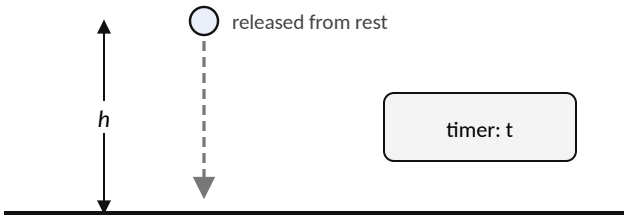
(a) The crate speeds up uniformly from rest to 1.2 m/s upward in 2.0 s. Calculate the tension in the cable during this time.

(b) The crate then rises at a constant 1.2 m/s. Calculate the tension in the cable during this time.

(c) Finally, the crate slows uniformly from 1.2 m/s to a stop in 1.5 s while still moving up. Calculate the tension in the cable during this time.

(d) The cable can safely hold a tension of at most 5.0×10^3 N. Calculate the largest upward acceleration the crane can give this crate.

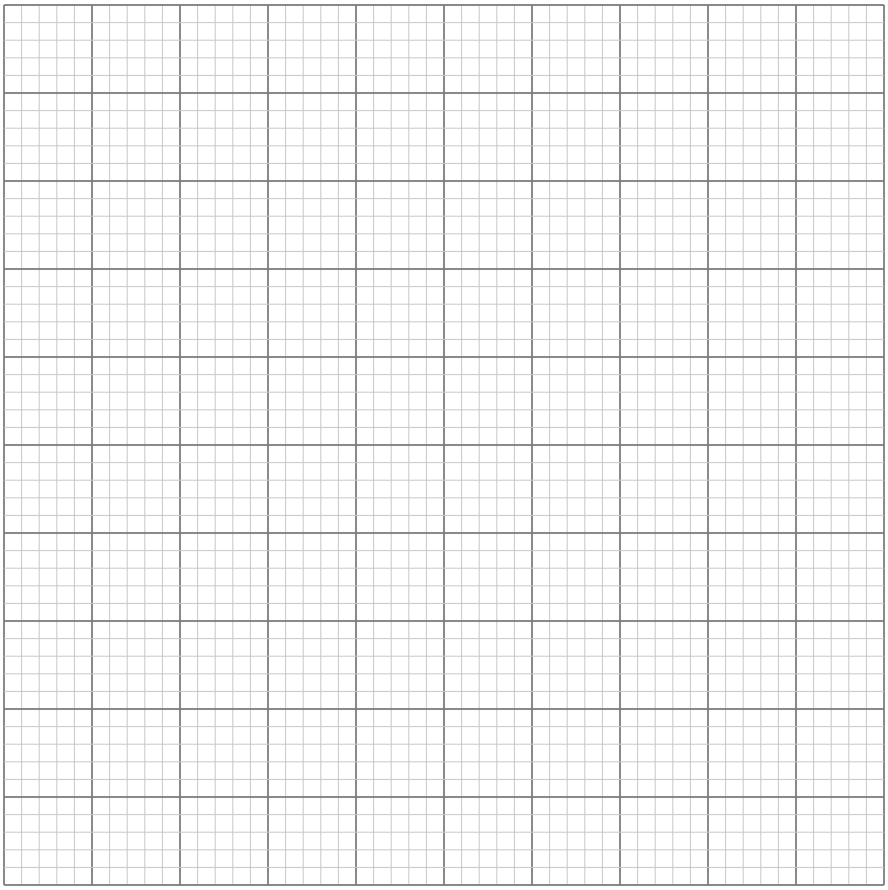
4. (25 points) Astronauts on an unknown planet drop a small ball from rest from several heights h and use a timer to measure the time t it takes to reach the ground. The planet has no air. For an object dropped from rest, $h = \frac{1}{2}gt^2$, where g is the planet's gravitational field strength. To get a straight line, they will graph t^2 on the vertical axis and h on the horizontal axis.



Height h (m)	Fall time t (s)	t^2 (s^2)
0.50	0.41	
1.00	0.58	
1.50	0.70	
2.00	0.83	
2.50	0.90	
3.00	1.01	

(a) Use the blank column in the table to calculate t^2 for each height.

(b) Plot t^2 (vertical) against h (horizontal) on the grid. Label each axis with the quantity and units, and choose a scale that uses more than half the grid.



(c) Draw the best-fit line. Calculate its slope, with units, from two far-apart points on the line. Show the two points you use.

(d) Starting from $h = \frac{1}{2}gt^2$, show that the slope of the line equals $2/g$. Then calculate g on this planet, with units, and state whether this planet's gravitational field is stronger or weaker than Earth's.

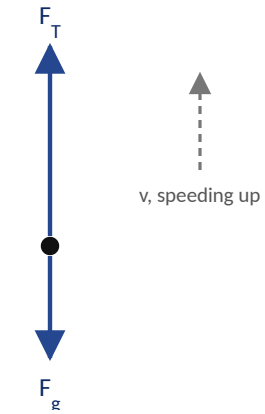
Answer Key – Practice Test, Version 4 – Introductory Dynamics

Worked step by step, the same way as the class notes: sketch, free-body diagram, $a = F_{net}/m$, forces with signs, algebra first, then numbers with units. $g = 9.8\ m/s^2$; answers match the significant figures of the givens.

Every problem: a = ΣF/m (equation sheet), written $a = F_{net}/m$. Replace F_{net} with the actual forces and their signs ($F_T - F_g$, $F_{th} - F_g$, $F_N - F_g$), swap $F_g = mg$, solve in letters, then plug in.
g is the gravitational field strength: $g = F_g/m$, in N/kg (1 N/kg = 1 m/s^2).
Kinematics, the first three on the sheet: #1 $v = v_0 + at$ · #2 $\Delta x = v_0t + \frac{1}{2}at^2$ · #3 $v^2 = v_0^2 + 2a\Delta x$.

1. Free-body diagrams (25 points)

(a) crate



Step 1 List every force on the crate.

Earth pulls down (F_g). The cable pulls up (F_T). Nothing else touches the crate.

Step 2 Check the lengths.

Moving up and speeding up means a points up, so F_{net} points up: $F_T > F_g$. The F_T arrow is longer.

Watch out: there is no separate "upward motion" force. The tension is the only force pulling up.

(b) lamp, at rest

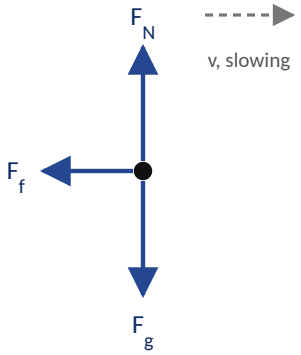


Step 3 At rest, so a = 0.

$F_{net} = 0$, so $F_T = F_g$: equal arrows. The cord pulls up and Earth pulls down.

Watch out: the lamp pulls down on the cord, but that force acts on the cord, so it does not go on the lamp's diagram.

(c) car, braking



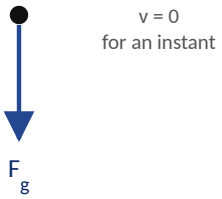
Step 4 Vertical, then horizontal.

No vertical acceleration, so $F_N = F_g$ (equal arrows).

Braking: friction F_f from the road points left, opposite the motion. It is the only horizontal force, so F_{net} and a point left.

Watch out: no arrow to the right. Moving right does not need a force to the right; the car keeps going because of inertia.

(d) ball, at the very top



Step 5 **Only gravity.**

Once the ball leaves the hand, nothing touches it and air resistance is negligible, so the only force is F_g, down.

Step 6 **(d) Why there is no upward force.**

A force needs something to exert it, and nothing is touching the ball to push it up. **It rose because of its inertia, not a force**

Gravity slowed it the whole way up; at the top v = 0 for an instant, but F_g still acts, so a = g down.

Watch out: v = 0 at the top does not mean a = 0 or F_{net} = 0.

2. g as field strength, and a lander (25 points)

Step 1 **(a) Field strength is weight per kilogram.**

$$g = \frac{F_g}{m} = \frac{55.7\ \text{N}}{15.0\ \text{kg}} = 3.713\ \text{N/kg} \rightarrow \boxed{g_{Mars} = 3.71\ \text{N/kg}}$$

Three significant figures, matching 55.7 N and 15.0 kg. It is about 38% of Earth's 9.8 N/kg.

Step 2 **(b) Mass does not depend on location.**

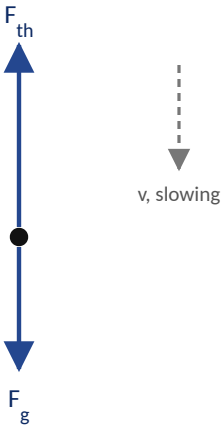
m = 15.0 kg, the same as on Earth Mass measures how much matter (inertia) the rover has, and moving it to Mars does not change that.

Only the weight F_g = mg changes, because g changes.

Step 3 **(c) Direction of the acceleration first. Up is +.**

velocity	speed	acceleration
down	decreasing (slowing)	opposite v: up, a = +1.50 m/s²

(c) lander



Two forces: Mars pulls down (F_g), and the exhaust gas pushes up (F_{th}). The acceleration is up, so the F_{th} arrow is longer.

Step 4 **Newton's second law, in letters.**

$$a = \frac{F_{net}}{m} \rightarrow a = \frac{F_{th} - F_g}{m} \rightarrow F_{th} - mg = ma \rightarrow F_{th} = m(g + a)$$

Step 5 **Now the numbers, with g_{Mars} from (a).**

$F_{th} = (6.20 \times 10^2\ \text{kg})(3.71\ \text{m/s}^2 + 1.50\ \text{m/s}^2)$

$= (6.20 \times 10^2\ \text{kg})(5.21\ \text{m/s}^2) = 3230\ \text{N} \rightarrow$ **3.23×10^3 N, upward**

Three significant figures, matching 6.20×10^2 kg, 1.50 m/s² and 3.71 N/kg.

Watch out: slowing down while moving down means the acceleration is up. Using a = −1.50 m/s² gives 1.37×10^3 N, less than the weight, which would make the lander speed up.

Watch out: use g for Mars (3.71 N/kg), not 9.8 N/kg. The lander is not on Earth.

Step 6 **(d) Third-law partner.**

$F_{A \text{ on } B} = -F_{B \text{ on } A}$. The thrust is $F_{\text{gas on lander}}$, so its partner is $\boxed{F_{\text{lander on gas}} = 3.23 \times 10^3 \text{ N, downward}}$

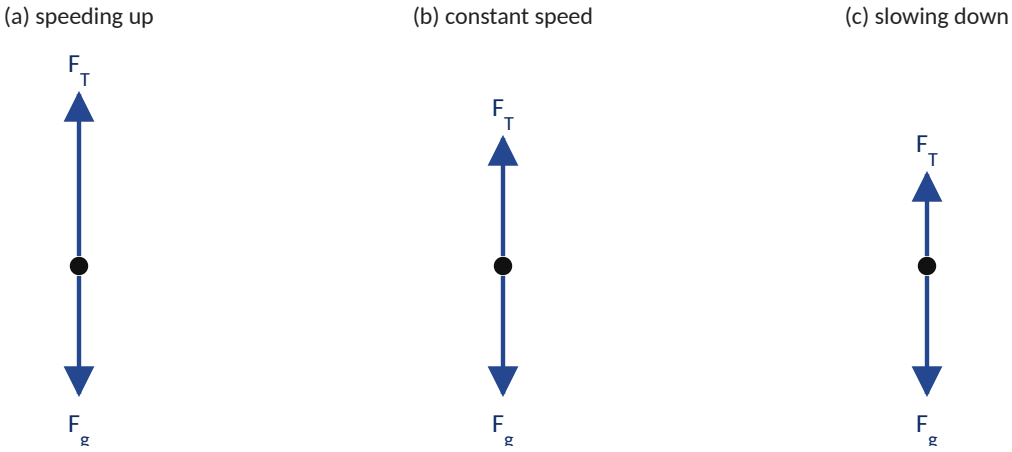
The lander pushes the exhaust gas down; the gas pushes the lander up. Same magnitude, opposite directions, on different objects.

3. Crane and crate: tension in three stages (25 points)

Step 1 **One equation for every stage. Up is +.**

$$a = \frac{F_{net}}{m} \rightarrow a = \frac{F_T - F_g}{m} \rightarrow F_T - mg = ma \rightarrow F_T = m(g + a)$$

Only a changes from stage to stage, so find a first, then plug in.



The weight is the same in every stage; only the tension changes. The crate moves up in all three.

Step 2 **(a) Speeding up: acceleration from the change in velocity (#1).**

$$a = \frac{v - v_0}{t} = \frac{1.2 \text{ m/s} - 0}{2.0 \text{ s}} = 0.60 \text{ m/s}^2 \text{ (up)}$$

$F_T = m(g + a) = (450 \text{ kg})(9.8 \text{ m/s}^2 + 0.60 \text{ m/s}^2) = (450 \text{ kg})(10.4 \text{ m/s}^2) = 4680 \text{ N} \rightarrow \boxed{4.7 \times 10^3 \text{ N}}$

Step 3 **(b) Constant speed: a = 0.**

$F_T = m(g + 0) = mg = (450 \text{ kg})(9.8 \text{ m/s}^2) = 4410 \text{ N} \rightarrow \boxed{4.4 \times 10^3 \text{ N}}$

Equal to the weight: no net force is needed to keep moving at constant velocity.

Step 4 **(c) Slowing while moving up: a points down.**

$$a = \frac{v - v_0}{t} = \frac{0 - 1.2 \text{ m/s}}{1.5 \text{ s}} = -0.80 \text{ m/s}^2$$

$F_T = m(g + a) = (450 \text{ kg})(9.8 \text{ m/s}^2 + (-0.80 \text{ m/s}^2)) = (450 \text{ kg})(9.0 \text{ m/s}^2) = 4050 \text{ N} \rightarrow \boxed{4.1 \times 10^3 \text{ N}}$

Less than the weight, even though the crate is still rising.

Watch out: the crate moves up in all three stages, but the tension is larger than the weight only while it speeds up. The sign of a, not the direction of v, decides.

Step 5 **(d) Largest acceleration: solve the same equation for a, in letters first.**

$F_T - mg = ma \rightarrow a = \dfrac{F_T - mg}{m} = \dfrac{F_T}{m} - g$

$a_{max} = \dfrac{5.0 \times 10^3\ \text{N}}{450\ \text{kg}} - 9.8\ \text{m/s}^2 = 11.1\ \text{m/s}^2 - 9.8\ \text{m/s}^2 = 1.31\ \text{m/s}^2 \rightarrow$ **1.3 m/s², upward**

Two significant figures, matching 5.0×10^3 N.

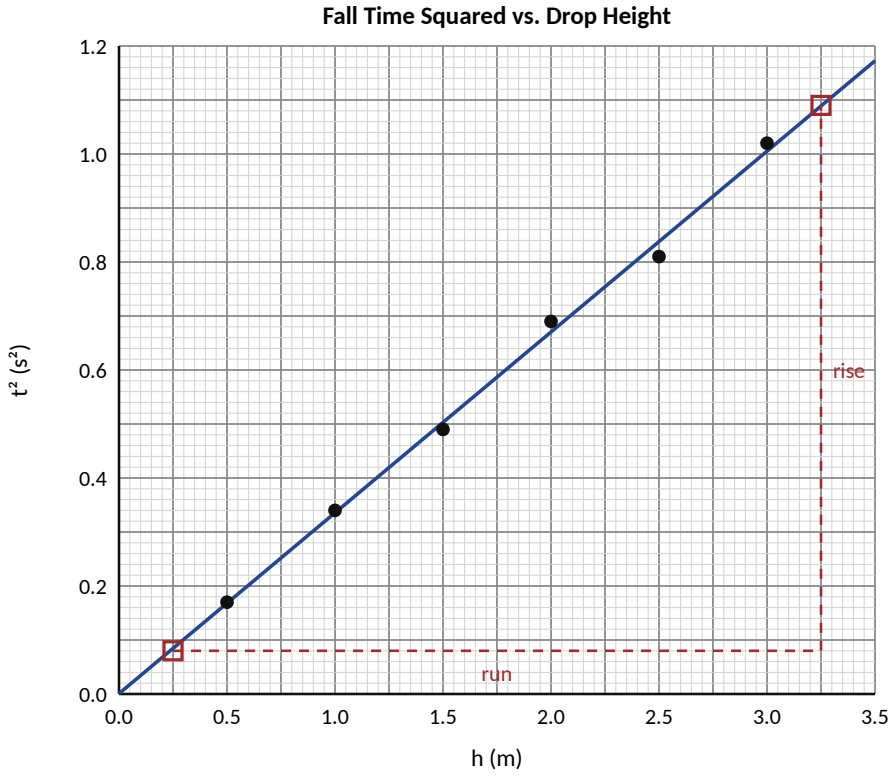
4. Linearize the data (25 points)

Step 1 **(a) Square each time; the units square too (s²).**

Example: $(0.41\ \text{s})^2 = 0.1681\ \text{s}^2 \rightarrow 0.17\ \text{s}^2$. Round to two decimal places, like the times.

h (m)	t (s)	t² (s²)
0.50	0.41	0.17
1.00	0.58	0.34
1.50	0.70	0.49
2.00	0.83	0.69
2.50	0.90	0.81
3.00	1.01	1.02

Step 2 **(b) Labeled axes with units, a scale over half the grid; (c) one straight line through the middle.**



The red squares are points read off the line, not data points.

Step 3 **(c) Slope from two far-apart points on the line, units carried.**

Points on the line: (0.25 m, 0.08 s^2) and (3.25 m, 1.09 s^2)

$$\text{slope} = \frac{1.09\ s^2 - 0.08\ s^2}{3.25\ m - 0.25\ m} = \frac{1.01\ s^2}{3.00\ m} = 0.337\ s^2/m$$

slope ≈ 0.34 s^2/m Any value from 0.31 s^2/m to 0.36 s^2/m from a reasonable line earns credit.

Step 4 **(d) Solve $h = \frac{1}{2}gt^2$ for the vertical quantity, t^2, and match it to $y = mx + b$.**

$$h = \tfrac{1}{2}gt^2 \rightarrow t^2 = \frac{2h}{g} \rightarrow t^2 = \frac{2}{g} \cdot h + 0$$

vertical (y)	horizontal (x)	slope	intercept
t^2 (s^2)	h (m)	2/g	0

$$\text{slope} = \frac{2}{g} \rightarrow g = \frac{2}{\text{slope}} = \frac{2}{0.337\ s^2/m} = 5.94\ m/s^2 \rightarrow$$ **g ≈ 5.9 m/s^2 (5.9 N/kg)**

Units: 2 ÷ (s^2/m) = m/s^2. Any value from 5.6 m/s^2 to 6.5 m/s^2 earns credit, consistent with the slope range.

Weaker than Earth's 5.9 N/kg is less than 9.8 N/kg, so the same object weighs less there.

Watch out: g = 2/slope, not 1/slope and not 2 × slope. The ½ in $h = \frac{1}{2}gt^2$ is where the 2 comes from.

1. Free-body diagrams

Suggested partial credit	pts
(a) Tension up and gravity down, each starting on the dot; labeled	3
(a) Tension arrow longer than gravity (speeding up while rising)	3
(b) Tension up and gravity down, equal lengths	4
(c) Normal force up and gravity down, equal lengths	3
(c) Friction to the left, and no force to the right	4
(d) Gravity only, pointing down	4
(d) Nothing touches the ball to push it up; it rose because of inertia	4
Total	**25**

2. g as field strength, and a lander

Suggested partial credit	pts
(a) $g = F_g/m$ written in letters, then numbers with units	2
(a) 3.71 N/kg	3
(b) 15.0 kg	2
(b) Mass does not depend on location; only the weight mg changes	3
(c) Thrust up, longer than gravity down; labeled	4
(c) a = +1.50 m/s^2 (up), because it slows while moving down	3
(c) $F_{th} - mg = ma$ solved for F_{th} in letters	3
(c) 3.23×10^3 N, units in every parenthesis, using g on Mars	2
(d) Lander on the exhaust gas, 3.23×10^3 N, downward	3
Total	**25**

3. Crane and crate

Suggested partial credit	pts
(a) $a = \Delta v/\Delta t = 0.60\ m/s^2$	3
(a) $F_T - mg = ma$ solved for $F_T = m(g + a)$ in letters	3
(a) 4.7×10^3 N, units in every parenthesis	2
(b) Constant speed, so $a = 0$ and $F_T = mg$	3
(b) 4.4×10^3 N	2
(c) $a = -0.80\ m/s^2$ (slowing while moving up)	3
(c) Same equation with the negative a	2
(c) 4.1×10^3 N, less than the weight	2
(d) $a = F_T/m - g$ solved in letters first	3
(d) $1.3\ m/s^2$, upward	2
Total	**25**

4. Linearize the data

Suggested partial credit	pts
(a) t^2 column correct, in s^2	3
(b) t^2 on the vertical axis, h on the horizontal axis, each with units	3
(b) Scale uses more than half the grid	2
(b) Points plotted correctly	2
(c) One straight best-fit line, not forced through the origin	2
(c) Two far-apart points read off the line, not data points	2
(c) Slope with units, between $0.31\ s^2/m$ and $0.36\ s^2/m$	3
(d) $t^2 = (2/g)h$ from $h = \frac{1}{2}gt^2$, so the slope is 2/g	3
(d) $g = 2/\text{slope}$ with units, between $5.6\ m/s^2$ and $6.5\ m/s^2$	3
(d) Weaker than Earth's, because g is less than 9.8 N/kg	2
Total	**25**